

Experiment-7

Program to Implement BFS

Aim

To develop a Python program to implement the Breadth-First Search (BFS) algorithm for graph traversal.

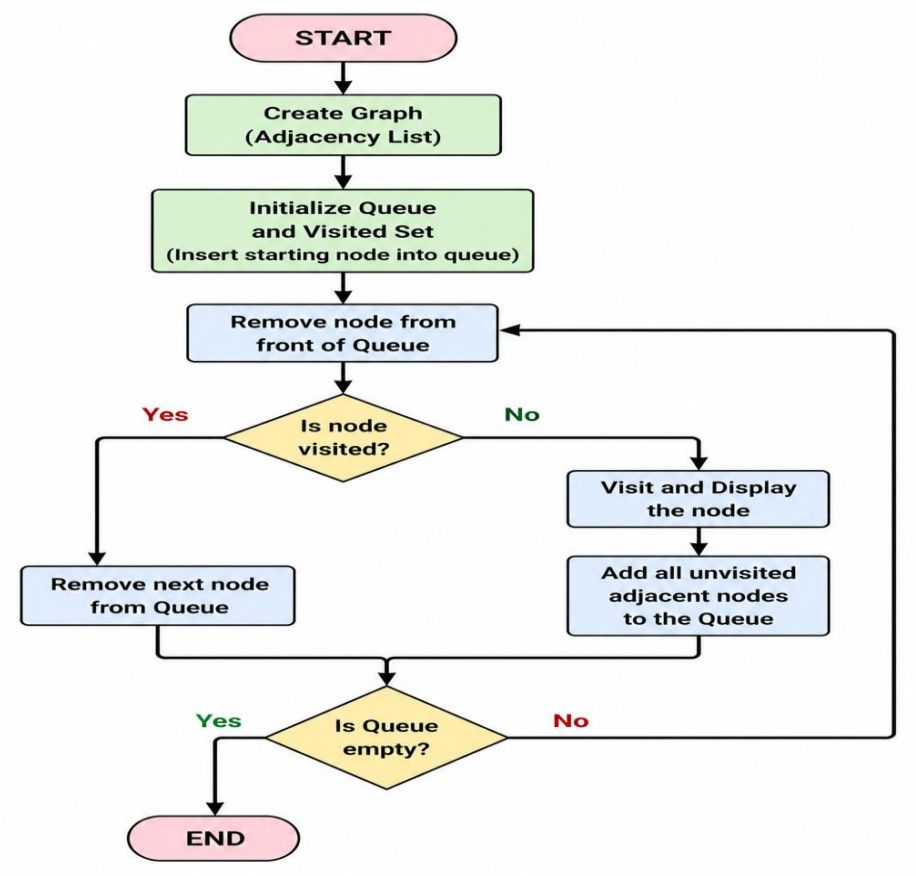
Objective

- To implement the Breadth-First Search (BFS) algorithm.
- To traverse all nodes of a graph level by level.
- To understand graph traversal techniques in Artificial Intelligence.

Algorithm

1. Start the program.
2. Create the graph using an adjacency list.
3. Initialize an empty visited set.
4. Insert the starting node into a queue.
5. Remove the front node from the queue.
6. If the node is not visited, mark it as visited and display it.
7. Add all unvisited adjacent nodes to the queue.
8. Repeat until the queue becomes empty.
9. Stop the program.

Flowchart:



Code:

```
from collections import deque
```

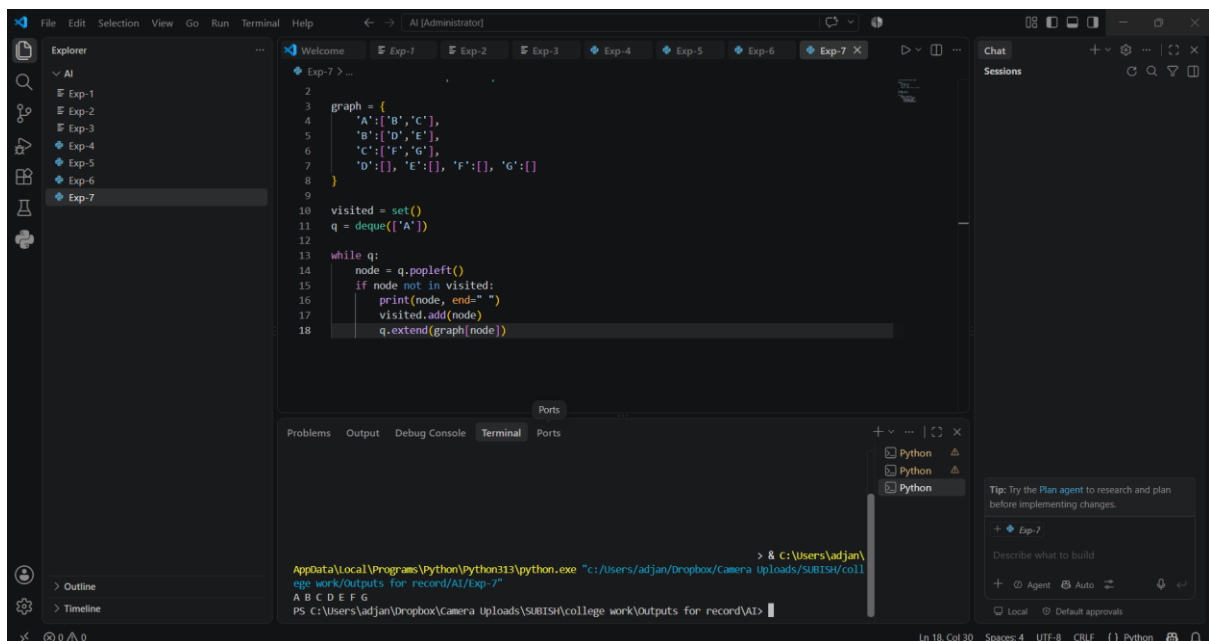
```
graph = {  
    'A':['B','C'],  
    'B':['D','E'],  
    'C':['F','G'],  
    'D':[], 'E':[], 'F':[], 'G':[]  
}
```

```
visited = set()  
q = deque(['A'])
```

```
while q:
```

```
node = q.popleft()
if node not in visited:
    print(node, end=" ")
    visited.add(node)
    q.extend(graph[node])
```

Output:



The screenshot shows a VS Code editor with a Python script implementing Breadth-First Search (BFS). The script defines a graph with nodes A through F and their neighbors. It uses a queue and a visited set to traverse the graph level by level. The terminal output shows the nodes being visited in the order A B C D E F G.

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [], 'E': [], 'F': [], 'G': []
}

visited = set()
q = deque(['A'])

while q:
    node = q.popleft()
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        q.extend(graph[node])
```

Terminal Output:

```
> & C:\Users\adjan\
AppData\Local\Programs\Python\Python313\python.exe "c:/Users/adjan/Dropbox/Camera Uploads/SUBISH/college work/Outputs for record/AI/Exp-7"
A B C D E F G
PS C:\Users\adjan\Dropbox\Camera Uploads\SUBISH\college work\Outputs for record\AI>
```

Result:

The Breadth-First Search (BFS) algorithm was successfully implemented, and all graph nodes were traversed in level-order.

Conclusion:

The Breadth-First Search (BFS) algorithm efficiently traverses a graph level by level using a queue, ensuring that all reachable nodes are visited systematically.